1

Statistics and Data Science Collaboration Laboratories: Engines for Development

Eric A. Vance
University of Colorado Boulder

Tonya R. Pruitt
Virginia Tech

CONTENTS

1.1 Introduction: Why Statistics and Data Science Have Extraordinary Potential for Data-Driven Development 4
1.2 Collaborative Statisticians and Data Scientists 5
1.3 Statistics and Data Science Collaboration Laboratories ("Stat Labs") 7
1.3.1 The Purpose of Stat Labs 7
1.3.2 What Stat Labs Do 8
1.3.2.1 Supporting Domain Experts 8
1.3.2.2 Creating New Knowledge 9
1.3.2.3 Transforming Evidence into Action (TEA) 10
1.3.2.4 Training the Next Generation of Collaborative Statisticians and Data Scientists 11
1.3.3 Stat Labs Produce Collaborative Statisticians and Data Scientists and Data-Capable Development Actors 11
1.4 Exemplar Stat Labs 12
1.4.1 The Laboratory for Interdisciplinary Statistical Analysis (LISA) at Virginia Tech 12
1.4.2 LISA at the University of Colorado Boulder 14
1.4.3 The University of Ibadan LISA (UI-LISA) 16
1.5 Stat Labs Can Become Engines for Development 17
1.5.1 Theory of Change 17
1.5.2 Benefits and Impacts of a Stat Lab 18
1.6 Seven Steps for Creating a Stat Lab 18
1.7 What Makes a Stat Lab Strong and Sustainable 20
1.8 Conclusion 22
References 22

DOI: 10.1201/9781003261148-2

1.1 Introduction: Why Statistics and Data Science Have Extraordinary Potential for Data-Driven Development

Statistical analysis and data science have enormous potential for enabling and accelerating data-driven development. The discipline of statistics has been around for more than 100 years, and its impacts are widespread and profound. Statistics has been driving development since even before the world's first university statistics department was founded in 1911 at University College London (Hotelling 1988). For example, during the great cholera outbreak in London in 1854, John Snow collected and analyzed data to identify and close off a contaminated water pump (Tulchinsky 2018). In the same year, Florence Nightingale, the founder of modern nursing and a very early member of the International Statistical Institute and the American Statistical Association, used statistics and data visualization to show that simple hygiene measures could drastically reduce infection and death in hospitals (McDonald 1998). As Gunderman and Vance (2021) wrote, the ability to collect and learn from large amounts of data has been a major driver of innovation over recent decades. Everything from health care—including patient analytics (Adebanji et al. 2015; Awe et al. 2021; Kosorok and Laber 2019), wearable devices (Michie et al. 2017), and the COVID-19 response (Aidoo et al. 2021; Gayawan et al. 2020)—to energy (Malakar et al. 2021) to entertainment recommender systems (e.g., Netflix (Bell et al. 2007)), is now driven by data and statistics.

We consider data science to be the science of learning from data (Donoho 2017), which encompasses—at a minimum—statistics, computation, and ways of thinking about data. Clearly, the idea of learning from data is not new. The scientific method is based upon deriving and testing theories from empirical observation, i.e., data. Figure 1.1 shows how the wheel of statistics and data science turns through the scientific process to ultimately apply statistics and data science to solve problems and make decisions for the benefit of society. Starting with an understanding of domain experts' research, business, or policy questions, statisticians and data scientists collaborate with domain experts to design an experiment or study to produce high-quality data, or, alternatively, design an algorithm to collect or scrape already existing data.

Statisticians can furthermore collaborate with data producers to collect high-quality data (Vance and Love 2021). All statisticians and data scientists have the responsibility to understand—and reduce when possible—the sources of variation and potential biases within the data. Statisticians and data scientists visualize the data and analyze data sets with models and algorithms to produce findings relevant to the original research, business, or policy questions. Statistics can help with interpreting the practical and statistical significance of results leading to conclusions about the data and a greater understanding of the underlying questions. This, in turn, may lead to collecting and analyzing more data in another turn of the wheel, in time progressing to solving the problem, making recommendations, and making decisions. The final step for a statistician or data scientist who wants to make a deep contribution is to collaborate with others to implement data-driven solutions, recommendations, and decisions to take action for the benefit of society.

Later in this chapter, Love et al. (2022) ask and answer why we need to develop statistical practice and collaboration in developing countries. In short, local capacity in statistics and data science can enable and accelerate innovative, sustainable solutions to local development challenges. When local statisticians and data scientists collaborate with local domain experts to transform evidence into action—especially when it occurs within a statistics

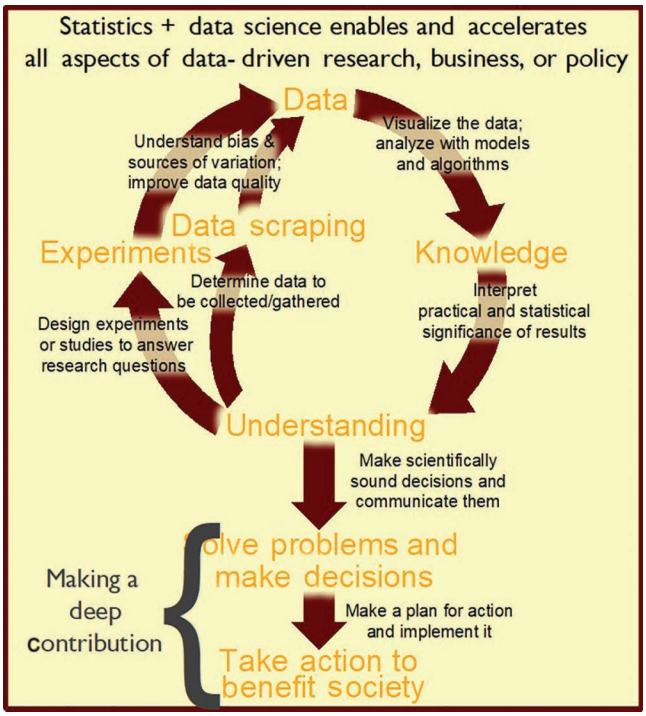


FIGURE 1.1
Statistics and data science enables and accelerates all aspects of data-driven research, business, or policy.

and data science collaboration laboratory—a virtuous cycle of development emerges in which initial collaborations build the capacity of all parties to achieve even more positive development outcomes by applying statistics and data science.

The goal of this chapter is to explain the concept of the statistics and data science collaboration laboratory ("stat lab") and how it can sustainably harness the extraordinary potential of statistics and data science for development. In Section 1.2, we introduce the generator of a stat lab—the collaborative statistician or data scientist. In Section 1.3, we explain what a stat lab is and what it does. We provide three examples of stat labs in Section 1.4. In Section 1.5, we discuss how stat labs can become engines for development and attempt to answer the question: Why should a university create and sustain a stat lab? We walk through the seven steps to create a stat lab in Section 1.6. Based on the lessons learned so far from the LISA 2020 Network, in Section 1.7, we characterize what makes stat labs strong and sustainable. We conclude in Section 1.8.

1.2 Collaborative Statisticians and Data Scientists

A technically trained statistician, who understands the theory and methods of statistics, is well positioned to do many things in academia, business and industry, or the policy domain. One option for the academic statistician is to teach the theory and methods of

statistics to others and/or continue to develop new theory and methods. Another path available to all statisticians is to apply the theory and methods of statistics to solve problems that will help benefit society. For a statistician in business, it may be straightforward to identify problems that directly affect companies' products or profits (Oliveira et al. 2016). In academia, on the other hand, it can be a challenge to identify or be exposed to problems that have real-world relevance. In addition, universities, especially in developing countries, may fail to prepare statisticians with appropriate applied statistics skills, including the ability to compute and code effectively (Awe 2012; Love et al. 2022).

Drawing upon a definition proposed by Halvorsen et al. (2020), we define a *collaborative statistician or data scientist* to be

> someone possessing deep technical skills in the theory and methods of statistics or data science and effective collaboration skills who can move between theory and practice to work with domain experts to create solutions to research, business, and policy challenges and achieve research, business, and policy goals.

In other words, a collaborative statistician or data scientist can transform evidence (data) into action, i.e., the solving of research problems or the making of data-driven business and policy decisions.

Fortunately, we have found that those with a strong background in theory and methods of statistics can learn applied and collaborative skills while working on consulting projects and more involved collaborative projects within a statistical collaboration laboratory (Vance et al. 2016). Through mentorship from an experienced individual or via a formal program, a statistician can become a *collaborative statistician or data scientist*; collaborate with domain experts on problems in research, business, or policy (see Figure 1.2); and subsequently have a more positive impact on society (Love et al. 2017; Vance et al. 2017a, 2017b). An individual can also learn collaboration skills through self-study of Vance and Smith's ASCCR (Attitude–Structure–Content–Communication–Relationship) Framework for Collaboration (2019).

Working alone, a collaborative statistician or data scientist can enable and accelerate as many as ten projects or more per year. For example, collaborative statisticians can help understand the policy implications of uneven distribution of costly medical interventions (Vance et al. 2013), impacts of climate change on disease prevalence in Botswana (Alexander et al. 2013), the cost benefits of upgrading domestic water systems in Senegal (Hall et al. 2015a), and the productive uses of piped water in Africa (Hall et al. 2014b).

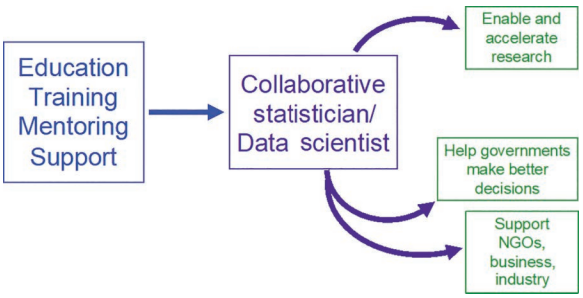


FIGURE 1.2
A well-trained collaborative statistician or data scientist can enable and accelerate many research, business, and policy projects.

Yet another path awaits a collaborative statistician or data scientist, one with orders of magnitude more potential impact. A collaborative statistician or data scientist can create a statistics and data science collaboration laboratory.

1.3 Statistics and Data Science Collaboration Laboratories ("Stat Labs")

A statistics and data science collaboration laboratory, or "stat lab," is a collection of statisticians and data scientists who have been trained to collaborate with domain experts and then do so to enable and accelerate research and make data-driven business and policy decisions. Similar to a statistical consulting center or a university "Center of Excellence," a stat lab is where members of the community can go to access statistics and data science expertise. Stat labs are not "rooms full of computers." Instead, they are more like rooms full of collaborative statisticians and data scientists working with domain experts from multiple disciplines. The term stat lab refers to both the human resources and the physical space they occupy. The statisticians and data scientists who work in stat labs all have different relative strengths—like the comic book and movie superheroes "The Avengers"—and likewise work together to achieve a common purpose: to enable and accelerate data-driven development while training the next generation of collaborative statisticians and data scientists.

1.3.1 The Purpose of Stat Labs

Stat labs can have different missions based on the host institution's statistics and data science needs. Vance and Laga (2017) found in a survey of North American stat labs that their primary mission was (given in order) serving clients and keeping them happy, training students, advancing research, improving clients' statistical skills, generating funds, and advancing the statistical methodology. While a few stat labs' primary mission was to generate funds for their university or department, this potential mission was, overall, the least popular and least relevant for the majority of stat labs, likely because stat labs within universities are not businesses. Rather, their purpose is to educate—not to make money. Successful stat labs that do generate a profit reinvest those funds to support the lab's mission, i.e., building more capacity. If a stat lab were primarily motivated by profit, it could become a private business off campus, or its director/coordinator could become a private statistical consultant.

Stat labs in the LISA 2020 Network (see Part 7) have a threefold mission:

1. Training the next generation of collaborative statisticians and data scientists to move between theory and practice to solve problems for real-world impact
2. Supporting research, innovation, and the transformation of evidence to action through collaboration with domain experts by providing necessary statistical analysis and data science expertise (i.e., create infrastructure for collaboration)
3. Improving statistical skills and data literacy widely by teaching short courses and workshops (i.e., upskill data-capable development actors)

1.3.2 What Stat Labs Do

At a fundamental level, stat labs do four things: they support researchers and data decision-makers, they create new knowledge, they transform knowledge or evidence into action, and they do all of this while building capacity for data-driven development by training the next generation of collaborative statisticians and data scientists.

1.3.2.1 Supporting Domain Experts

Support for the clients or *domain experts* utilizing the stat lab takes several forms. Stat labs teach short courses and workshops to improve statistical analysis capacity and data literacy widely.

Members of stat labs listen to researchers, businesses, and policymakers and provide advice and consultations, and often provide emotional support for their work. When these researchers, businesspeople, and policymakers are not sure about some aspect of their study design, data analysis, or interpretation of data, they can ask an expert in the stat lab. Often that statistician or data scientist may not be able to answer immediately but can find the answer by referencing books or articles, asking a colleague, or searching the internet for a solution.

When the statistician or data scientist goes beyond teaching, advising, or consulting and contributes some aspect to the final result or product of the project, we call this *collaboration* (Love et al. 2017). Examples of collaboration include designing an experiment or study; performing power calculations or determining sample sizes; analyzing data; writing a summary of the statistical methods used; creating plots or graphs to explore and communicate data; interpreting results; communicating findings, conclusions, and recommendations.

One of the three missions of LISA 2020 stat labs is to improve the statistical skills and data literacy of members of their community (e.g., students in other departments, faculty, and staff across the university or institution, local school children, nonuniversity community members). This can occur during the collaboration process. As the domain expert teaches the statistician or data scientist relevant information about their domain, so does the statistician or data scientist teach the domain expert about statistics or data science. Vance et al. (2022a) expatiate on how to create shared understanding between collaborators, both about domain issues and statistics and data science technical issues.

The primary way in which stat labs improve the statistical skills and data literacy of members of their community is by teaching short courses or workshops. These short courses and workshops can be on specific topics the stat lab thinks many people should know, such as how to design an experiment or how to design and analyze a survey. The topics can also be driven by demand from domain experts. For example, if several domain experts come to the stat lab with similar problems or questions about statistics—such as how to conduct an ANOVA—the stat lab could teach a 2-hour short course on that topic so that attendees can learn how to do it themselves. Sometimes a department will request the stat lab to teach a 1- or 2-day workshop such as "Statistical Methods for Biology." For all short courses and workshops, the goal of the stat lab is to improve the statistics and data literacy of the participants. Ideally, this will lead to the participants becoming good users of statistics and data science. At the very least, participants should become more aware of the power of statistics and data science, learn what methods may enable or accelerate their research or decision-making, and recognize when they should collaborate with the stat lab for needed expertise.

1.3.2.2 Creating New Knowledge

Stat labs create new knowledge in at least four ways.

1. **Stat labs help researchers answer questions they could not have answered without expert statistical advice**: Two examples are collaborations in which multiple statisticians from a stat lab helped water and sanitation specialists improve pit latrine management in low-income areas of Malawi (Chirwa et al. 2017) and understand the timing of the infancy-childhood growth transition in rural Gambia (Bernstein et al. 2020). Another example is of a study of agriculturally focused study abroad programs in Asia (Sharma et al. 2014).
2. **Based on an understanding of the domain experts' goals and their data, stat labs may suggest novel questions the data can answer**: In one case, statisticians analyzing data from a water project recognized that the data could answer the question of how willing households were to pay for enhanced toilet systems in Senegal (Hall et al. 2015b). Another example is when statisticians used data from an impact evaluation of a rural water supply project in Mozambique (Hall et al. 2014a) in collaboration with a social scientist to better understand the intracommunity impacts of that project (Van Houweling et al. 2017). In both cases, the research teams deemed the novel questions raised by the statisticians to be very important and impactful, and the statisticians became authors on the papers, leading to the discovery and dissemination of new knowledge.
3. **When stat labs encounter new types of data for which standard statistical methods do not apply, they can create new knowledge by developing novel methods that enable researchers to extract useful information from their data**: In collaboration with a professor of African history, three stat lab statisticians developed a novel integration of spatial statistical models with a Markov decision process to determine the most likely movement paths of people enslaved during the collapse of the Oyo Empire in Africa in the 1820s (Weins et al. 2022). Another example is a novel process of cleaning data during data collection developed by a statistician working on a project in Mozambique (Seiss et al. 2014). Developing new statistical methods to answer research questions arising from stat lab collaborations can be a core component of PhD dissertations; see Vance et al. (2009) and Vance (2008) for an example of this.
4. **Stat labs can generalize their lessons learned to improve statistical practice and collaboration for all**: Based on more than 40 years of combined experience collaborating with domain experts, running stat labs, and training more than 280 collaborative statisticians and data scientists, Vance and Smith (2019) created a framework to learn and teach interdisciplinary collaboration. Additional examples of knowledge creation about collaboration includes an article on goals for statistics and data science collaborations (Vance 2020), how to assess statistical consultations and collaborations (Vance et al. 2020), and—elsewhere in this volume—learning and teaching the ABC of collaboration (Awe and Vance 2022) and using the TEAM framework for collaboration (Adenomon 2022b). With such a rich experience collaborating in different contexts to enable and accelerate data-driven development, more knowledge about best practices in statistical practice and collaboration is sure to be produced by stat labs in the future. In fact, all of the chapters in this volume are examples of knowledge produced by stat labs.

1.3.2.3 Transforming Evidence into Action (TEA)

Stat labs aspire to transform evidence into action (TEA), for what good is knowledge created (i.e., academic evidence) if it leads nowhere and fails to make a positive impact on society? Olubusoye et al. (2021, p. 13) describe the logic of TEA:

> If we can create and sustain stat labs comprised of well-trained statisticians and data scientists who can move between theory and practice to apply statistics to solve problems and make decisions, and if they can collaborate with development actors empowered to produce data and take action on development issues, then the members of the stat labs can transform data into evidence to answer their collaborators' research questions or help them make data-driven policy decisions that will lead to development impacts and benefits to society.

TEA projects are collaborations between stat labs and *data decision-makers*, who are development actors working in a variety of settings: government officials who may be creating or implementing policy, employees of non-governmental organizations who may also be creating or implementing policy, businesses or entrepreneurs making decisions that will ultimately affect how many people they hire or what goods and services they produce, or researchers who drive innovation and discovery based on results of their experiments or studies. The end goal of a TEA project is a positive development impact based on a data-driven innovation.

The following are the reasons described by Vance and Love (2021) why TEA projects are difficult to implement and fully realize:

- Statisticians and data scientists must be skilled collaborators as well as skilled methodologists and analysts. To achieve TEA they must be able to understand the data and projects they are working with on both a deep and broad level. They must also and be able to communicate the results of statistical methods and analytical work in ways that provide actionable evidence to those who can use it to positively impact society.
- Policymakers may not know of the existence of a stat lab; therefore, stat labs must deliberately reach out to them. By reaching out early as skilled collaborators, stat labs can ensure that they are helping to answer questions of interest to local policy or business decision-makers.
- TEA requires a mindset to see a project all the way to its end. Policy change is often a long and complex process; stat labs may feel compelled to move on to a new project before the impacts of the old one are realized. Nevertheless, to transform evidence into action for the benefit of society, statisticians and data scientists must adopt a TEA mindset and have the patience to engage in the complete process.
- Measuring longer term impacts of stat lab activities and describing the contribution of statistics and data science is challenging. So even if a TEA project resulted in a positive development impact, the stat lab may not know about it, or others may not know of the impact the stat lab had on the project's outcome.

However, TEA is worth aspiring toward. Olubusoye et al. (2021, p. 13) write:

> TEA projects have tremendous potential to enable and accelerate data-driven development because they operate in a space occupied by three development actors: data

producers, data decision makers, and data analyzers. Stat labs provide collaborative statistics and data science expertise to bring all of the actors together to produce evidence and transform it into action for development.

1.3.2.4 Training the Next Generation of Collaborative Statisticians and Data Scientists

A second terminal goal of TEA projects is to build the capacity of the stat lab to have even more development impact in the future. In fact, the primary purpose of a stat lab is to train students and staff to become effective interdisciplinary collaborators who can move between theory and practice to solve problems for real-world impact.

Vance et al. (2022a) describe an educational paradigm in which statistics and data science students learn collaboration through a five-stage process: preparation, practice, doing, reflecting, and mentoring. As part of their training to work in a stat lab, students prepare by reading articles and observing senior stat lab members collaborate with domain experts, they practice aspects of the ASCCR Framework for collaboration in the classroom (Vance and Smith 2019), they then perform collaboration by working on a stat lab project under the mentorship of senior collaborators, they reflect on what went well and what could be improved, and finally, they reach the stage where they can mentor others in collaboration.

This education and training process requires the involvement of the stat lab to provide students with opportunities to observe (prepare), do, and mentor others on real projects. The other stages can occur within a classroom setting, such as in a capstone course for undergraduates or a statistical consulting and collaboration practicum course (Kolaczyk et al. 2021). Alternatively, the stat lab can conduct weekly staff meetings during which members discuss projects, thereby providing opportunities for students to practice collaboration skills and reflect on how to improve them.

Thus, a stat lab can provide all five stages of the process for education and training in collaboration.

An impactful way to help students progress through the stages of preparation, practice, reflecting, and mentoring is for stat labs to conduct Video Coaching and Feedback Sessions (VCFSs) in which a collaboration project meeting is video recorded (with permission of the domain expert) on a phone, webcam, or video camera and then reviewed in a small group of 4–7 stat lab members. Vance (2014) describes key features of VCFS, which include reviewing only a few (2–4) short clips of the meeting (only 1–5 minutes long), having observers focus on specific behaviors in the short video clips, conducting the VCFS with intention to help the participants improve their collaboration skills, having each participant share one lesson learned from the VCFS, and documenting lessons learned.

Much more is written about educating and training the next generation of collaborative statisticians and data scientists in Vance (2021) and in this edited volume (Gayawan et al. 2022; Nunes 2022). How stat labs can continue to drive such education and training in the practice of statistics will be a focus of continuing research.

1.3.3 Stat Labs Produce Collaborative Statisticians and Data Scientists and Data-Capable Development Actors

Like a laboratory that produces chemicals or invents a new technology, a statistics and data science collaboration laboratory ("stat lab") also produces outputs. We have discussed how stat labs create new knowledge and transform that into action for development. However, the primary output for a stat lab is human capacity in statistical analysis and data science; stat labs produce collaborative statisticians and data scientists and data-capable

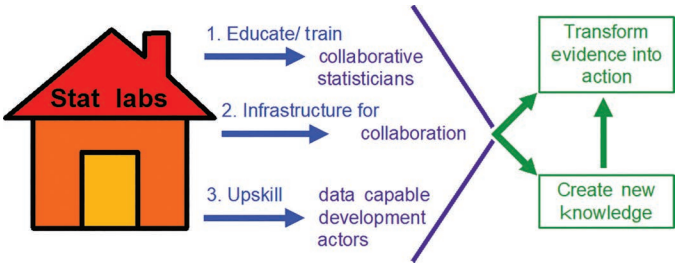


FIGURE 1.3
The activities of stat labs lead to the production of collaborative statisticians and data scientists, data-capable development actors, and space for them to collaborate to transform evidence into action. This leads to data-driven innovations and positive development impacts.

development actors. In sum, stat labs build capacity for data-driven development (Vance and Love 2021).

Figure 1.3 shows a diagram of how stat labs leverage their activities to produce collaborative statisticians and data scientists and data-capable development actors to build capacity for data-driven development. From this capacity, stat labs create knowledge and transform evidence into action, leading to development outcomes and impacts discussed in Section 1.5.

1.4 Exemplar Stat Labs

In an invited talk at the 62nd World Statistics Congress of the International Statistical Institute in 2019, Vance described seven components of a stat lab using an analogy to the human body (see Figure 1.4). Each stat lab has seven "body" parts. The heart is the mission/goals/objectives of the lab. The backbone of the stat lab is its administration. The muscles are the personnel and how they are trained. The head/brain are the stat lab's services. The legs are the lab's communication strategy. Its blood is the support it receives from its university and institution and its budget. And finally, its skin is the (visible) outcomes and impacts of the stat lab. In this section, we describe these seven components for two instantiations of the Laboratory for Interdisciplinary Statistical Analysis (LISA) at Virginia Tech and the University of Colorado Boulder. We also provide an example of a third stat lab, the University of Ibadan's LISA (UI-LISA).

1.4.1 The Laboratory for Interdisciplinary Statistical Analysis (LISA) at Virginia Tech

LISA was created at Virginia Tech in Blacksburg, Virginia, United States in 2008 within the Department of Statistics and lasted for 8 years until 2016 (Vance and Pruitt 2016b). It was built on the foundation of the Statistical Consulting Center (1973–2007) and the Statistical Laboratory (1948–1972), which preceded the establishment of the Department of Statistics in 1949 by 1 year (Arnold et al. 2013). LISA's mission was to train statisticians to become interdisciplinary collaborators, provide research infrastructure to enable and accelerate high-impact research, and engage with the Virginia Tech community in outreach activities to improve statistical skills and literacy (Vance and Pruitt 2016a).



FIGURE 1.4
Stat labs have seven essential components, analogous to the human body.

LISA provided three main services: collaboration on research projects, Walk-in Consulting to answer quick statistics questions, and short courses to teach new statistical methods. During the 2015–2016 fall and spring semesters, which was LISA's last year of operation at Virginia Tech, the 50 statistical collaborators of LISA met with researchers for individual statistical collaboration meetings on 294 projects. During daily Walk-in Consulting hours, LISA met with 468 faculty, staff, and students to answer quick statistical questions on projects requiring less than 30 minutes of assistance. Fifteen LISA Short Courses were offered to teach 455 graduate students and other university members how to apply statistics in their research. Overall, LISA provided at least 4,701 hours of statistical assistance and education to members of the Virginia Tech community. LISA's statistical collaborators reported 3,750.5 hours for statistical collaboration projects, 650.5 hours for Walk-in Consulting, and 300 hours for Short Courses. The number of hours reported worked in 2015–2016 fall and spring semesters is likely an underestimation of the total hours (Vance and Pruitt 2016b). By counting "Other" responses as half "Yes" and half "No", 91% of LISA clients in the academic year 2014–2015 considered LISA's services "helpful," and 90% were satisfied with their overall LISA experience. See Figure 1.5.

The bulk of LISA's workforce was MS and PhD students from the Department of Statistics. From 2008 to 2016, 223 students and six visiting scholars were educated and trained to move between theory and practice to help researchers apply statistics to answer

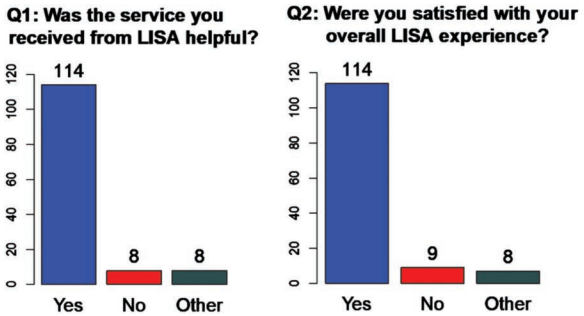


FIGURE 1.5
Summary of quantitative feedback on LISA collaboration meetings in 2014–2015.

their research questions (Vance et al. 2016). During the 2015–2016 fall and spring semesters, 50 students collaborated with researchers on LISA projects. These students had taken a semester-long course in communication and collaboration skills (Vance 2012, 2013) to prepare them for LISA and received continuing training while working in LISA via weekly staff meetings to discuss projects. Additionally, 41 of the students had at least one video recorded meeting with a domain expert reviewed during a VCFS (Vance and Pruitt 2016b). During each semester, four students had a fully funded teaching assistant position (50% time) to work in LISA. Also, a dozen other students each semester had a part-time (25% time) appointment in LISA. The remaining students were volunteers.

LISA was led by a full-time Director, Dr. Eric Vance, throughout its existence at Virginia Tech. From 2010 to 2016, its Assistant Director was Dr. Chris Franck, who was primarily supported financially by external research projects. From 2009 to 2016, Tonya Pruitt was LISA's Administrative Specialist. Her appointment in LISA was for 80% of her time. During the 2015–2016 fall and spring semesters, LISA hired two research associates to collaborate on long-term funded projects. The Director of LISA reported directly to the Statistics Head of Department as well as to the Vice President of Research and the Dean of the College of Science. Having strong, dedicated personnel contributed immensely to the success of LISA.

Statistical assistance from LISA was free for Virginia Tech faculty, staff, and students. LISA was funded jointly by the Office of the Vice President of Research, the College of Science, the Graduate School, the Office of the Provost, and all seven other colleges: Agriculture and Life Sciences, Architecture and Urban Studies, Engineering, Liberal Arts and Human Sciences, Natural Resources and Environment, the Pamplin College of Business, and the Virginia-Maryland Regional College of Veterinary Medicine. The Department of Statistics provided funding for a dozen 25%-time LISA statistical collaborators each semester. It also provided physical space and other infrastructural support such as internet access, accounting, printing, and computers for LISA's activities. The total annual budget for LISA from its joint funders was $300,000.

LISA disseminated its annual reports to college deans and other university administrators. It advertised its services mainly through flyers around campus and word of mouth. However, its short courses were advertised in weekly email newsletters sent by the Graduate School to graduate students, and these short courses always communicated the collaboration and Walk-in Consulting services of LISA. LISA was featured in several national statistics publications (Carzolio 2014; Crandell 2015; Stallings 2014) and was featured in six consecutive invited sessions of the Joint Statistical Meetings (JSM) from 2011 to 2016.

LISA's primary outcomes were the 229 students and visiting fellows it trained to become collaborative statisticians and the thousands of clients who benefited from LISA's services. There were 2,719 total LISA collaboration projects, 2,962 Walk-in Consulting visitors, and 3,748 short course attendees. At least 80 papers stemming from LISA projects were authored by LISA collaborators and published in high-impact journals, with at least 49 of them coauthored by students in LISA. LISA faculty were on 40 grant-funded projects.

1.4.2 LISA at the University of Colorado Boulder

LISA moved to the Department of Applied Mathematics at the University of Colorado Boulder (CU Boulder) in 2016 when its Director, Prof. Eric Vance, took a position there as an Associate Professor. Because the local context of CU Boulder was very different from

that of Virginia Tech, LISA transformed itself into a new lab while retaining many of the characteristics that previously made it strong. LISA's current goal is to increase the quantity and quality of statistics and data science applied to advance high-impact research at CU Boulder and expert decision-making in the community. Its threefold mission is to train statisticians and data scientists to become interdisciplinary collaborators, provide research infrastructure to enable and accelerate quantitative research around the campus community, and engage with the community to improve statistical skills and data literacy (Vance 2017).

LISA serves as research infrastructure by collaborating with researchers on campus on big and small projects—both externally funded and unfunded—that lead to a variety of outcomes such as coauthored publications, grant proposals, internal reports, preliminary data for future projects, theses and dissertations, etc. From 2016 to 2021, LISA has averaged around 80 such collaboration projects annually. LISA also hosts a research service called "LISA Statistics and Data Science Walk-in/Zoom-in Hours" and has averaged around 70 walk-in (or Zoom-in during the pandemic) visitors. LISA has taught an average of 15 short courses and workshops per year since 2017 to an average of 200 attendees annually. The quantitative feedback from domain experts collaborating with LISA from 2016 to 2021 was remarkably similar to the feedback for LISA in the academic year 2014–2015: 92% considered LISA's services helpful, and 90% were satisfied with their overall LISA experience.

LISA trains statisticians and data scientists to communicate and collaborate with researchers to solve research problems and work with businesses and government agencies to make data-driven decisions. LISA statistical collaborators can help design experiments, studies, and surveys; collect, analyze, and plot data; run statistical software; interpret results; and develop strategies to communicate statistical concepts and results to nonstatisticians. LISA has trained 111 graduate and undergraduate students, two postdoctoral researchers, and one visiting fellow in statistics and data science collaboration principles through a three-credit course on statistical collaboration (Vance and Smith 2019). Each student collaborates on three LISA projects via this course.

Twenty students have taken an advanced course in statistical collaboration during which they engaged in VCFS and collaborated on an additional five to six LISA projects. Students discuss projects during weekly staff meetings as part of both courses.

LISA's Director remains Prof. Vance. Dr. Judith Law taught the LISA collaboration course for two semesters in 2020–2021. A current grant from the National Science Foundation funds administrative support for tracking the projects and their outcomes. LISA had temporary support from the College of Arts and Sciences to hire two graduate students per semester (50% time) to collaborate with researchers, mentor junior students, host Walk-in Hours, and teach short courses. LISA is currently self-sufficiently funded from grant projects and consulting contracts with a budget of approximately $100,000 per year. CU Boulder and the Department of Applied Mathematics provide physical space for collaboration meetings and Walk-in Hours as well as general information technology and accounting support and internet access.

Because the supply of students who work in LISA comes through a course on statistical collaboration, LISA advertises only as much as it needs to so that each student can work on three projects. Most advertisement of LISA's services occurs through its website (www.colorado.edu/lab/lisa), direct emails to campus departments, and word of mouth. LISA's short courses are advertised on campus by the Center for Research Data and Digital Scholarship. LISA's annual reports are sent to the chair of the Department of Applied Mathematics and deans in the College of Arts and Sciences.

LISA's primary outcomes were the 111 students, two postdocs, and one visiting fellow it trained to become collaborative statisticians and the hundreds of domain experts who have benefited from LISA's services. At least 24 papers stemming from LISA projects were authored by LISA collaborators and published in high-impact journals, with at least 18 of them coauthored by students in LISA. An additional 16 manuscripts are under review in high-impact journals. LISA was written into 11 grant-funded projects.

The mission, goals, objectives, services, training, and communication strategy of LISA at Virginia Tech (2008–2016) were nearly the same as that of LISA at CU Boulder (2016–present). However, when LISA moved to CU Boulder, it had to adapt to its new conditions. A primary difference is that around 90% of the students involved in LISA at Virginia Tech were MS and PhD students from the Department of Statistics, whereas the students of LISA at CU Boulder are much more diverse. More than half of the students of LISA at CU Boulder are undergraduates and around one-quarter of the LISA students have come from departments other than Applied Mathematics. Only 20% of the students have been MS or PhD students from the Department of Applied Mathematics. In fact, the number of MS or PhD students in a statistics or data science program of study at CU Boulder is about one-tenth as large as at Virginia Tech. The Director of the two LISAs has remained the same, but there was considerably more supporting staff and financial support at Virginia Tech. Additionally, at Virginia Tech, the Director was able to devote his efforts full-time to LISA, a luxury few stat labs and their directors can afford. In contrast—and like most other stat lab coordinators or directors—at CU Boulder Dr. Vance has the regular responsibilities of a standard faculty member, i.e., he has additional teaching, research, and service leadership responsibilities on top of his LISA duties. Consequently, LISA at Virginia Tech had more impacts and outcomes.

1.4.3 The University of Ibadan LISA (UI-LISA)

The University of Ibadan LISA (UI-LISA) was established in 2015 as the third member of the LISA 2020 Network through the influence of O. Olawale Awe, the First LISA Fellow. The primary objective of UI-LISA is to train statistics students to be able to employ the power of statistics to solve societal problems, become effective interdisciplinary collaborators, demonstrate the value of statistical thinking, and be excellent statistical communicators. UI-LISA has designed, tested, and implemented several programs, two of which were unique. These innovative programs include: (1) one-hour with a statistician, where basic and popularly used statistical tools and methods are introduced and explained to a diverse audience; (2) walk-in consultation where domain experts are provided with instant solutions to their statistical problems; (3) providing short statistical courses designed to improve statistical literacy and skills of all users of statistics at all levels; (4) a mobile statistical clinic where the UI-LISA team stations itself at public locations such as halls of residence, conference areas, and recreational areas to provide on-the-spot solutions to problems and inquiries related to statistics; and (5) engaging in statistical collaboration with development actors, including non-governmental and governmental agencies. Much more detail about this exemplar stat lab is provided in Olubusoye et al. (2022) and Adepoju et al. (2022).

Examples of additional stat labs are provided elsewhere in this chapter (Mawora et al. 2022), in Chapter 2 (Adenomon 2022a), as well as in Msemo and Vance (2015), Goshu (2016), Amin and Vance (2016), Vivacqua et al. (2018), and Esterhuizen et al. (2021).

1.5 Stat Labs Can Become Engines for Development

1.5.1 Theory of Change

Stat labs dedicated to the threefold mission of training the next generation of collaborative statisticians and data scientists; supporting research, innovation, and the transformation of evidence to action; and improving statistical skills and data literacy widely in their communities will have begun the process of building their capacity for data-driven development. If these stat labs also have an appropriate administrative structure, adequate training of their staff and students to provide services consistent with their mission, an effective communication strategy, and sufficient support from their university or institution, then they will be well positioned to engage in a positive feedback loop in which projects they collaborate on lead to innovations in development and positive development outcomes, the strengthening of the lab's capacity to collaborate on such projects, and consequently more opportunities and projects to make a positive impact on society (see Figure 1.6).

This positive cycle occurs because stat labs that reach out to the local community of researchers, business leaders, government agencies, and non-governmental organizations to provide both training and customized statistical support cultivate data-capable development actors (Vance and Love 2021). The community becomes more aware of the necessity of statistics and data science expertise, more capable in its use, and presents the lab with more requests for collaboration. New projects provide greater opportunities for the lab to train its students and staff and develop its own capacity to support data-driven development.

Stat labs aspire to be agents for change in their university or institution, surrounding community, and country. Stat labs can create a culture of interdisciplinary collaboration to transform evidence into action for development. A key difference between stat labs and more traditional statistical consulting centers is the stat lab's emphasis on education and training, i.e., building capacity to make data-driven decisions (Vance 2015). By focusing on the training of their own personnel and domain experts in the community to transform evidence into action, stat labs can unlock the potential of statisticians, data scientists, and domain experts alike. In short, stat labs can become engines for driving sustainable development.

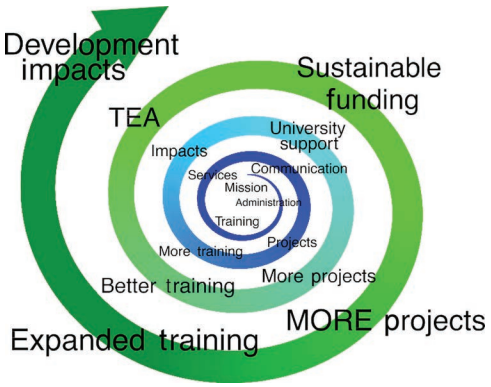


FIGURE 1.6
The virtuous cycle for a stat lab to become strong and sustainable.

1.5.2 Benefits and Impacts of a Stat Lab

We believe that there are many benefits that accrue from creating a stat lab. Stat labs provide *enhanced visibility* for the work of collaborative statisticians and data scientists. Many statisticians already collaborate with domain experts informally. Creating a stat lab elevates this work and provides a mechanism for statisticians to gain formal credit for their collaborations.

Stat labs also make the work of collaborative statisticians and data scientists more *accessible*. Creating a centralized place to go for statistics and data science expertise opens research opportunities to a more diverse set of researchers, businesses, and policymakers.

For university administrators (Vice-Chancellors, Deans, et al.) who aspire to lead their institutions toward a higher profile and a greater positive impact on society, the benefits of creating and sustaining a stat lab are numerous. Stat labs can help foster a culture of interdisciplinary collaboration at the university. Immediately following from this is that stat labs enable more academic staff to conduct high-quality research. A widespread gap in research is an inability to use—or ignorance of—appropriate statistical methods for designing, analyzing, and describing one's research (Vance et al. 2022b). Furthermore, appropriate statistical analyses can enable researchers to publish in high-impact journals. More academic staff publishing research in high-impact journals can lead to an improved reputation for the university, as well as a greater societal impact. A related benefit is for post-graduate students writing theses and dissertations. Assistance with the statistics and data science required for their research improves the quality of their research and shortens time to graduation.

At a lower administrative level, stat labs can help deans and heads of departments accomplish their goals for distinguishing their departments of statistics or mathematics as leaders in statistics and data science education and research. The opportunity for statistics/mathematics students to gain experience applying their statistics and data science skills on real projects positively impacts their education. Students with experience in stat labs report learning statistical theory better by applying it. They also gain practical skills working in a stat lab highly useful and sought after in the job market (Vance et al. 2016). Departments with stat labs can also connect with a global network of stat labs via the LISA 2020 Network (Vance et al. 2022b). These connections can help guide curricular development and global, collaborative research.

Table 1.1 contrasts potential outcomes stemming from creating and sustaining a stat lab with the vision laid out in this paper to what may be the current, status-quo reality at most universities that either have no stat lab or have a statistical center that does not operate like a stat lab.

1.6 Seven Steps for Creating a Stat Lab

We have developed a seven-step process for creating a new stat lab and becoming a member of the LISA 2020 Network. This process is described in Table 1.2. Step 5—opening the lab to train students and provide services—usually occurs after labs have completed the preparatory steps 1–4. However, the steps could be taken out of order. For example, a stat lab could open before they have completed the full stat lab plan/proposal of step 3, though we do not recommend it. The full stat lab plan/proposal available at www.lisa2020.org/

TABLE 1.1

Difference between the Status Quo and the Extraordinary Potential of Stat Labs

The Status Quo: Consulting Centers	What Could Be: Stat Labs
Staff engage in consulting	Staff and students engage in collaboration
Consulting on projects after data are collected	Collaborations with domain experts through all phases of research
Statistics is equations and proofs on a chalkboard (theory only)	Collaborative statisticians use data to solve real-world problems
	Provides research infrastructure
	Statistics drives a laboratory for creating knowledge
Minimal training of students	Students learn by doing
	Students trained by experienced mentors
Guessing what happened with clients	Analyzing video data on what really happened
Only academic staff consult	Students get jobs based on collaboration experience
Dreams of international travel	Students and staff travel internationally to do statistics and data science
Students struggle with statistics	Statistics becomes a strength for the university
Students take too long to complete their MSc and PhD	Collaboration with statisticians speeds time to graduation
Statistics is a barrier to research	Statistics enables and accelerates research at the university
Academic staff do not publish as many articles as they could	Staff publish in high-impact journals
	Staff are promoted
	University gains prestige
	University meets SOCIETAL NEEDS
	Become leading statistical collaboration laboratories
Department/center is isolated on campus	Stat lab connects department to research and educational initiatives across campus
Consulting center is isolated	Stat lab is connected to a global network

about/processes-and-policies guides potential labs through exercises to think about and plan for aligning the stat lab's mission with its institution's mission, identifying the stakeholders in the lab and the services that would make the lab a success to them, determining the lab's personnel and how they will be trained, establishing the lab's administrative structure, considering the budgetary needs of the lab and what support its institution will provide, communicating with the lab's stakeholders, and documenting the lab's outcomes and impacts. When a lab connects with the LISA 2020 Network (usually at step 1 or potentially at step 6 if their mentor is not from the network), the LISA 2020 Secretariat forms a committee to review and provide feedback on the proposed lab's plan/proposal.

Labs that identify a mentor from the LISA 2020 Network and complete steps 1–3 are considered "Proposed Members" of the network. Labs successfully responding to the feedback on the full lab plan/proposal (step 4) become "Transitional Members" of the network. The typical next step is to begin operation of the stat lab (step 5) and stay connected with the LISA 2020 Network (step 6). The final steps required to become a "Full Member" of the LISA 2020 Network are to report a full quarter of metrics and introduce the lab at a semimonthly LISA 2020 Network online Zoom meeting. A two-thirds majority vote of current full members is required to become a "Full Member." Detailed instructions are available at www.osf.io/2gqza/.

TABLE 1.2

Process for a Stat Lab to Become a Member of LISA 2020

Stat Lab Stages	Steps Required to Progress to the Next Stage
Potential	1. Identify the director/coordinator 1.1 Identify a mentor from within the LISA 2020 Network 2. Gather and document support 3. Complete the full lab plan/proposal
Proposed	3.1. Submit the full lab plan/proposal 4. Revise the proposal based on reviewer feedback
Transitional	4.1. Review committee approves revisions 5. Open the stat lab 5.1. Train students and staff 5.2. Provide research infrastructure 5.3. Teach short courses/workshops 5.4. Report activities, outcomes, and impacts (metrics) 6. Engage with the network 7. Submit a full quarter (3 months) of metrics 7.1. Give a formal presentation to introduce the stat lab to the network
Full	The Stat Lab's Full Lab Plan/Proposal, submitted metrics, and introduction presentation are sent to the LISA 2020 Network members for a final vote to determine if the transitional stat lab will be admitted as a full member. 2/3 Majority vote required to become a full member.

1.7 What Makes a Stat Lab Strong and Sustainable

In completing the seven steps described above, a new stat lab will have built a strong foundation. They will have secured initial departmental and university support. Their purpose, mission, goals, and objectives will provide a guiding light for lab members. Who will lead and manage the stat lab and participate as collaborators and what their roles and responsibilities will entail will have been established. The stat labs that have completed the process will have considered their unique institutional situation and determined which services they will provide. With this strong foundation—supplemented by feedback and recommendations received from their review panel—these new stat labs are ready to begin operation and build themselves into strong and sustainable engines for development.

The path to a strong and sustainable stat lab begins with training effective collaborative statisticians and data scientists who can move between theory and practice to support research, innovation, and the transformation of evidence into action within and beyond their institutions. This focus on training—especially to impart an evidence to action mindset—creates the conditions needed to transform academic evidence into data-driven innovations that lead to sustainable development impacts. Training, therefore, is the first step in a virtuous cycle where increased collaborations between statisticians and domain experts to develop data-driven innovations to solve local development challenges leads to strengthened stat labs with increased capacity to collect, organize, visualize, analyze, and present development-relevant data (see Figure 1.6). The more projects a stat lab works on, the more opportunities there are for senior statisticians to mentor junior statisticians, and the more quickly junior statisticians strengthen their capacity. These collaborations

also lead to enhanced capability of development actors to access and use statistical data for development and greater awareness of the power of statistics and data science to strengthen research and inform decision-making.

Initial successes create a positive feedback loop in which more researchers and development actors want to collaborate with the stat labs and become "data-capable," and more statistics students, faculty, and staff want to work in the stat lab. We have learned that successful projects beget many more requests from domain experts to collaborate with statisticians (Vance and Love 2021). Stat labs that focus on using projects as opportunities to train students rise to the challenge of building capacity quickly to successfully complete a greater number of projects.

When projects come with funding, senior students and faculty can be compensated to both work on projects and mentor junior students. Successful, high-profile projects may also engender support for stat labs from administrators within their institutions—potentially loosening restrictive rules and removing institutional barriers to success—and attract more students to study statistics and data science. This self-reinforcing cycle of using experience on projects to build capacity to work on more projects is a key to the sustainability of strong stat labs.

Effective personnel and a positive reputation are essential for a stat lab, but these alone are not enough to ensure the lab is strong and sustainable. Figure 1.4 is a diagram expressing the analogy of the essential components of a stat lab being like seven parts of a human body—the Heart is the Mission/Goals/Objectives; the Backbone is the Administration; the Muscles are the Personnel/Training; the Head/Brain are the Services; the Legs are Communication Strategy; the Blood is University Support/Budget; and the Skin is Impacts & Outcomes. As the health of each of these body parts is essential for a vigorous human body, the same can be said for a strong and sustainable stat lab. If any one part becomes weak, it impairs the whole system in the short term and can debilitate the stat lab for the long term if the issues are not addressed. Some issues can be addressed internally: ensuring the Mission/Goals/Objectives are integrated with those of the home department and institution, reviewing which services have been successful and which could be improved, exercising the muscles/training the collaborators, accurately keeping track of impacts and outcomes. Other issues may be more challenging and require working with those outside of the stat lab: insufficient operating funds, communicating the value of the stat lab, seeking additional collaboration opportunities. How to address each of these issues depends on the unique situation of the stat lab, but learning from the experiences and best practices of other stat labs—such as those in the LISA 2020 Network—provides direction and guidance. For example, stat labs struggling with statistical collaborator participation can explore incentivization methods that other network stat labs have implemented to determine which might work best at their institution.

In this respect, another fundamental aspect of sustainability is a connection to and being part of an organization larger than itself. Connections with lab mentors and the LISA 2020 Network provide encouragement, support, and guidance as the stat lab encounters both challenges and triumphs. The stat labs can share their experiences, lessons learned, and best practices to strengthen the individual labs and the network as a whole. Regular network meetings, symposia, and staff exchanges encourage communication and engagement between stat labs and enhance opportunities for collaboration. The impetus, development, and growth of the LISA 2020 Network is described in detail later in this chapter (Vance et al. 2022b).

1.8 Conclusion

The goal of this chapter is to explain how a collaborative statistician or data scientist who wants to have more positive impact on society can create a statistical collaboration laboratory, what such a stat lab does, and how stat labs can be created and sustained to become engines for development. We hope this article provides a useful guide for such collaborative statisticians or data scientists that can also be used to help convince academic staff and administrators to support the creation and sustainable activities of stat labs. When operated on the strong and sustainable path, a virtuous cycle of development emerges in which initial collaborations build the capacity of all parties to achieve even more positive development outcomes, enabling the stat lab to achieve its extraordinary potential to apply statistics and data science to positively impact humanity.

References

Adebanji, A., Adeyemi, S., and Gyamfi, M. (2015), "Empirical Analysis of Factors Associated with Neonatal Length of Stay in Sunyani, Ghana," *Journal of Public Health and Epidemiology, Academic Journals*, 7, 59–64. https://doi.org/10.5897/JPHE2014.0679.

Adenomon, M. O. (2022a), "Statistical Consulting and Collaboration Practices: The Experience of NSUK-LISA Stat Lab." *Promoting Statistical Practice and Collaboration in Developing Countries*, eds. O.O. Awe, K. Love, and E.A. Vance, Boca Raton, FL: CRC Press, pp. 131–138. DOI: 10.1201/9781003261148

Adenomon, M. O. (2022b), "Teaching and Learning Statistics in Nigeria with the Aid of Computing and Survey Data Sets from International Organization." *Promoting Statistical Practice and Collaboration in Developing Countries*, eds. O.O. Awe, K. Love, and E.A. Vance, Boca Raton, FL: CRC Press, pp. 371–383. DOI: 10.1201/9781003261148

Adepoju, A. A., Olubusoye, O. E., Ogundunmade, T., Tomiwa, O. T., and Robert, B. (2022), "The Complementary role of UI-LISA in Statistical Training and Capacity Building at the University of Ibadan, Nigeria." *Promoting Statistical Practice and Collaboration in Developing Countries*, eds. O.O. Awe, K. Love, and E.A. Vance, Boca Raton, FL: CRC Press, pp. 111–122. DOI: 10.1201/9781003261148

Aidoo, E. N., Ampofo, R. T., Awashie, G. E., Appiah, S. K., and Adebanji, A. O. (2021), "Modelling COVID-19 incidence in the African sub-region using smooth transition autoregressive model," *Modeling Earth Systems and Environment*. https://doi.org/10.1007/s40808-021-01136–1.

Alexander, K. A., Carzolio, M., Goodin, D., and Vance, E. (2013), "Climate Change is Likely to Worsen the Public Health Threat of Diarrheal Disease in Botswana," *International Journal of Environmental Research and Public Health*, 10, 1202–1230.

Amin, M., and Vance, E. A. (2016), "Inevitability of Interdisciplinary Statistical Laboratories for Food and Agricultural Research in Developing Countries: The case of Pakistan," in *Proceedings of ICAS VII Seventh International Conference on Agricultural Statistics*, Rome, Italy, pp. 1–8. https://doi.org/10.1481/icasVII.2016.h47b.

Arnold, J. C., Hinkelmann, K., Vining, G. G., and Smith, E. P. (2013), "Virginia Tech Department of Statistics," in *Strength in Numbers: The Rising of Academic Statistics Departments in the U. S.*, eds. A. Agresti and X.-L. Meng, New York, NY: Springer, pp. 537–546. https://doi.org/10.1007/978-1-4614-3649-2_39.

Awe, O. O. (2012), "Fostering the Practice and Teaching of Statistical Consulting among Young Statisticians in Africa," *European Journal of Business and Management*, 3, 39–44.

Awe, O. O., Dogbey, D. M., Sewpaul, R., Sekgala, D., and Dukhi, N. (2021), "Anaemia in Children and Adolescents: A Bibliometric Analysis of BRICS Countries (1990–2020)," *International Journal of Environmental Research and Public Health*, 18, 5756. https://doi.org/10.3390/ijerph18115756.

Awe, O. O., and Vance, E. A. (2022), "The ABC of Successful Statistical Collaborations: Adapting the ASCCR Framework in Developing Countries." *Promoting Statistical Practice and Collaboration in Developing Countries*, eds. O.O. Awe, K. Love, and E.A. Vance, Boca Raton, FL: CRC Press, pp. 149–156. DOI: 10.1201/9781003261148

Bell, R. M., Koren, Y., and Volinsky, C. (2007), "The Bellkor Solution to the Netflix Prize," *KorBell Team's Report to Netflix*.

Bernstein, R. M., O'Connor, G. K., Vance, E. A., Affara, N., Drammeh, S., Dunger, D. B., Faal, A., Ong, K. K., Sosseh, F., Prentice, A. M., and Moore, S. E. (2020), "Timing of the Infancy- Childhood Growth Transition in Rural Gambia," *Frontiers in Endocrinology*, 11(142). https://doi.org/10.3389/fendo.2020.00142.

Carzolio, M. (2014), "Masters Without Borders," *Amstat News*, 444, pp. 20–22.

Chirwa, C. F. C., Hall, R. P., Krometis, L.-A. H., Vance, E. A., Edwards, A., Guan, T., and Holm, R. H. (2017), "Pit Latrine Fecal Sludge Resistance Using a Dynamic Cone Penetrometer in Low Income Areas in Mzuzu City, Malawi," *International Journal of Environmental Research and Public Health*, 14, 87. https://doi.org/10.3390/ijerph14020087.

Crandell, I. (2015), "Cultural Values, Statistical Displays," *Amstat News*, 455, pp. 18–19.

Donoho, D. (2017), "50 Years of Data Science," *Journal of Computational and Graphical Statistics*, 26, 745–766. https://doi.org/10.1080/10618600.2017.1384734.

Esterhuizen, T. M., Li, G., Young, T., Zeng, J., Machekano, R., and Thabane, L. (2021), "Advancing Collaborations in Health Research and Clinical Trials in Sub-Saharan Africa: Development and Implementation of a Biostatistical Collaboration Module in the Masters in Biostatistics Program at Stellenbosch University," *Trials*, 22, 478. https://doi.org/10.1186/s13063-021-05427-x.

Gayawan, E., Awe, O. O., Oseni, B. M., Uzochukwu, I. C., Adekunle, A., Samuel, G., Eisen, D. P., and Adegboye, O. A. (2020), "The spatio-temporal epidemic dynamics of COVID-19 outbreak in Africa," *Epidemiology & Infection*, 148. https://doi.org/10.1017/S0950268820001983.

Gayawan, E., Somo-Aina, O., Aladeniyi, O. B., Owoeye, S. M., Aduloju, O. M., Adediran, A. A., and Orunmuluyi, O. S. (2022), "Technology and Multimedia in Statistical Education and Collaboration." *Promoting Statistical Practice and Collaboration in Developing Countries*, eds. O.O. Awe, K. Love, and E.A. Vance, Boca Raton, FL: CRC Press, pp. 515–556. DOI: 10.1201/9781003261148

Goshu, A. T. (2016), "Strengthening Statistics Graduate Programs with Statistical Collaboration—The Case of Hawassa University, Ethiopia," *International Journal of Higher Education*, 5, 217–221.

Gunderman, D., and Vance, E. (2021), "Low- and Middle-Income Countries Lack Access to Big Data Analysis—Here's How to Fill the Gap," *The Conversation*. https://theconversation.com/low-and-middle-income-countries-lack-access-to-big-data-analysis-heres-how-to-fill-the-gap-159412.

Hall, R. P., Davis, J., van Houweling, E., Vance, E. A., Seiss, M., and Russel, K. (2014a), *Impact Evaluation of the Mozambique Rural Water Supply Activity*, Blacksburg, VA: Virginia Tech School of Public and International Affairs.

Hall, R. P., Vance, E. A., and van Houweling, E. (2014b), "The Productive Use of Rural Piped Water in Senegal," *Water Alternatives*, 7, 480–498.

Hall, R. P., Vance, E. A., and van Houweling, E. (2015a), "Upgrading Domestic-Plus Systems in Rural Senegal: An Incremental Income-Cost (IC) Analysis," *Water Alternatives*, 8, 317–336.

Hall, R. P., Vance, E. A., van Houweling, E., and Huang, W. (2015b), "Willingness to Pay for VIP Latrines in Rural Senegal," *Journal of Water, Sanitation and Hygiene for Development*, 5, 586–593. http://dx.doi.org/10.2166/washdev.2015.053.

Halvorsen, K. T., Hanford, K. J., Vance, E. A., Wilson, J., and Zahn, D. (2020), "Transforming Your Stumbling Blocks into Stepping Stones," in *JSM Proceedings*, Alexandria, VA: American Statistical Association: Statistical Consulting Section, pp. 2523–2541.

Hotelling, H. (1988), "Golden Oldies: Classic Articles from the World of Statistics and Probability: The Place of Statistics in the University," *Statistical Science*, 3, 72–83. https://doi.org/10.1214/ss/1177013002.

Kolaczyk, E. D., Wright, H., and Yajima, M. (2021), "Statistics Practicum: Placing 'Practice' at the Center of Data Science Education," *Harvard Data Science Review*. https://doi.org/10.1162/99608f92.2d65fc70.

Kosorok, M. R., and Laber, E. B. (2019), "Precision Medicine," *Annual Review of Statistics and Its Application*, 6, 263–286. https://doi.org/10.1146/annurev-statistics-030718-105251.

Love, K., Awe, O. O., Gunderman, D. J., Druckenmiller, M., and Vance, E. A. (2022), "LISA 2020 Network Survey on Challenges and Opportunities for Statistical Practice and Collaboration in Developing Countries." *Promoting Statistical Practice and Collaboration in Developing Countries*, eds. O.O. Awe, K. Love, and E.A. Vance, Boca Raton, FL: CRC Press, pp. 47–60. DOI: 10.1201/9781003261148

Love, K., Vance, E. A., Harrell, F. E., Johnson, D. E., Kutner, M. H., Snee, R. D., and Zahn, D. (2017), "Developing a Career in the Practice of Statistics: The Mentor's Perspective," *The American Statistician*, 71(1), 38–46. https://doi.org/10.1080/00031305.2016.1255257.

Malakar, S., Goswami, S., Ganguli, B., Chakrabarti, A., Roy, S. S., Boopathi, K., and Rangaraj, A. G. (2021), "Designing a long short-term network for short-term forecasting of global horizontal irradiance," *SN Applied Sciences*, 3, 477. https://doi.org/10.1007/s42452-021-04421-x.

Mawora, T., Otieno, J., and Vance, E. A. (2022), "Exploring the Need for a Statistical Collaboration Laboratory in a Kenyan University: Experiences, Challenges, and Opportunities." *Promoting Statistical Practice and Collaboration in Developing Countries*, eds. O.O. Awe, K. Love, and E.A. Vance, Boca Raton, FL: CRC Press, pp. 61–70. DOI: 10.1201/9781003261148

McDonald, L. (1998), "Florence Nightingale: Passionate Statistician," *Journal of Holistic Nursing*, 16, 267–277. https://doi.org/10.1177/089801019801600215.

Michie, S., Yardley, L., West, R., Patrick, K., and Greaves, F. (2017), "Developing and Evaluating Digital Interventions to Promote Behavior Change in Health and Health Care: Recommendations Resulting from an International Workshop," *Journal of Medical Internet Research*, 19, e7126. https://doi.org/10.2196/jmir.7126.

Msemo, E., and Vance, E. A. (2015), "LISA 2020: Impacting Agricultural Productivity in Tanzania through the Wheels of Statistics," in *Proceedings of the International Statistical Institute's 60th World Statistics Congress*, Rio de Janeiro.

Nunes, M. A. (2022), "Modernizing the Curricula of Statistics Courses through Statistical Learning." *Promoting Statistical Practice and Collaboration in Developing Countries*, eds. O.O. Awe, K. Love, and E.A. Vance, Boca Raton, FL: CRC Press, pp. 351–362. DOI: 10.1201/9781003261148

Oliveira, T., Oliveira, A., Mahmoudvand, R., Ravishankar, N., and Banks, D. (2016), *Book of Abstracts ISBIS 2016: Meeting on Statistics in Business and Industry*, Lisbon: Universidade Aberta.

Olubusoye, O. E., Akintande, O. J., and Vance, E. A. (2021), "Transforming Evidence to Action: The Case of Election Participation in Nigeria," *CHANCE*, 34(3), 13–23.

Olubusoye, O. E., Alaba, O., Vance, E., Folorunso, S., and Akintande, O. J. (2022), "Promoting and Sustaining a Virile Statistical Laboratory in Nigeria's Premier University: Lesson from UI-LISA Experience." *Promoting Statistical Practice and Collaboration in Developing Countries*, eds. O.O. Awe, K. Love, and E.A. Vance, Boca Raton, FL: CRC Press, pp. 95–110. DOI: 10.1201/9781003261148

Seiss, M., Vance, E., and Hall, R. (2014), "The Importance of Cleaning Data During Fieldwork: Evidence from Mozambique," *Survey Practice*, 7(4), 2864.

Sharma, A., Marchant, M. A., Vance, E. A., Smith, E. P., Richardson, W. W., and Hightower, L.S. (2014), "National Survey of Study Abroad Programs Conducted in Asia Using the Food and Agriculture Education Information System (FAEIS) Database," *NACTA Journal*, 58, 142–149.

Stallings, J. (2014), "Type IV Errors: How Collaboration Can Lead to Simpler Analyses," *Amstat News*, 440, 24–25.

Tulchinsky, T. H. (2018), "Chapter 5—John Snow, Cholera, the Broad Street Pump; Waterborne Diseases Then and Now," in *Case Studies in Public Health*, ed. T. H. Tulchinsky, Academic Press, pp. 77–99. https://doi.org/10.1016/B978-0-12-804571-8.00017-2.

Van Houweling, E., Hall, R., Carzolio, M., and Vance, E. (2017), "'My Neighbour Drinks Clean Water, While I Continue To Suffer': An Analysis of the Intra-Community Impacts of a Rural Water Supply Project in Mozambique," *The Journal of Development Studies*, 53, 1147–1162. https://doi.org/10.1080/00220388.2016.1224852.

Vance, E. (2012), "Communication in Statistical Collaborations: Teaching Students How to Be Effective Interdisciplinary Collaborators," in *4th Annual Conference on Higher Education Pedagogy*, Blacksburg, VA: Virginia Tech's Center for Instructional Development and Educational Research, p. 233.

Vance, E. (2013), "Using Team-Based Learning to Teach Effective Communication and Collaboration," in *5th Annual Conference on Higher Education Pedagogy*, Blacksburg, VA: Virginia Tech's Center for Instructional Development and Educational Research, pp. 296–297.

Vance, E. A. (2008), "Statistical Methods for Dynamic Network Data," Ph.D., Durham, North Carolina: Duke University.

Vance, E. A. (2014), "LISA Video Coaching and Feedback Sessions," *Collaboration in a Bag*, p. 1. https://doi.org/10.17605/OSF.IO/XMTCE.

Vance, E. A. (2015), "Recent Developments and Their Implications for the Future of Academic Statistical Consulting Centers," *The American Statistician*, 69(2), 127–137. https://doi.org/10.1080/00031305.2015.1033990.

Vance, E. A. (2017), "LISA: Laboratory for Interdisciplinary Statistical Analysis 2016–17 Annual Report," *OSF*. https://doi.org/10.17605/OSF.IO/ZQEJ2.

Vance, E. A. (2020), "Goals for Statistics and Data Science Collaborations," in *JSM Proceedings*, Alexandria, VA: American Statistical Association: Statistical Consulting Section, pp. 2198–2209.

Vance, E. A. (2021), "Using Team-Based Learning to Teach Data Science," *Journal of Statistics and Data Science Education*, 29(3), 277–296. https://doi.org/10.1080/26939169.2021.1971587.

Vance, E. A., Alzen, J. L., and Seref, M. M. H. (2020), "Assessing Statistical Consultations and Collaborations," in *JSM Proceedings*, Alexandria, VA: American Statistical Association: Statistical Consulting Section, pp. 161–169.

Vance, E. A., Alzen, J. L., and Smith, H. S. (2022a), "Creating Shared Understanding in Statistics and Data Science Collaborations," *Journal of Statistics and Data Science Education*, 30(1), 54–64. https://www.tandfonline.com/doi/full/10.1080/26939169.2022.2035286

Vance, E. A., Archie, E. A., and Moss, C. J. (2009), "Social networks in African elephants," *Computational and Mathematical Organization Theory*, 15(4), 273–293. https://doi.org/10.1007/s10588-008-9045-z.

Vance, E. A., Glimp, D. R., Pieplow, N. D., Garrity, J. M., and Melbourne, B. (in press), "Integrating the Humanities into Data Science Education: Reimagining the Introductory Data Science Course," *Statistics Education Research Journal*.

Vance, E. A., LaLonde, D. E., and Zhang, L. (2017a), "The Big Tent for Statistics: Mentoring Required," *The American Statistician*, 71(1), 15–22. https://doi.org/10.1080/00031305.2016.1247016.

Vance, E. A., and Love, K. (2021), "Building Statistics and Data Science Capacity for Development," *CHANCE*, 34(3), 38–46.

Vance, E. A., Love, K., Awe, O. O., and Pruitt, T. R. (2022b), "LISA 2020: Promoting Statistical Practice and Collaboration in Developing Countries." *Promoting Statistical Practice and Collaboration in Developing Countries*, eds. O.O. Awe, K. Love, and E.A. Vance, Boca Raton, FL: CRC Press. DOI: 10.1201/9781003261148

Vance, E. A., Metzger, T., and Pruitt, T. (2016), "The Educational Impact of Working in a Statistical Collaboration Laboratory," in *Conference on Higher Education Pedagogy*, Blacksburg, VA.

Vance, E. A., and Smith, H. S. (2019), "The ASCCR Frame for Learning Essential Collaboration Skills," *Journal of Statistics Education*, 27(3), 265–274. https://doi.org/10.1080/10691898.2019.1687370.

Vance, E. A., Tanenbaum, E., Kaur, A., Otto, M. C., and Morris, R. (2017b), "An Eight-Step Guide to Creating and Sustaining a Mentoring Program," *The American Statistician*, 71(1), 23–29. https://doi.org/10.1080/00031305.2016.1251493.

Vance, E. A., Xie, X., Henry, A., Wernz, C., and Slonim, A. D. (2013), "Computed Tomography Scan Use Variation: Patient, Hospital, and Geographic Factors," *The American Journal of Managed Care*, 19, e93–e99.

Vance, E., and Laga, I. (2017), "Variations in Statistical Practice between North American Stat Labs," in *ASA Conference on Statistical Practice*, Jacksonville, FL.

Vance, E., and Pruitt, T. (2016a), *LISA: Virginia Tech's Laboratory for Interdisciplinary Statistical Analysis Annual Report 2014–15*, Report, Virginia Tech. Laboratory for Interdisciplinary Statistical Analysis.

Vance, E., and Pruitt, T. (2016b), *Virginia Tech's Laboratory for Interdisciplinary Statistical Analysis Annual Report 2015–16*, Report, Virginia Tech. Laboratory for Interdisciplinary Statistical Analysis.

Vivacqua, C. A., de Pinho, A. L. S., Nunes, M. A., and Vance, E. A. (2018), "Integrating Collaboration, Communication and Problem Solving to Promote Innovation in Statistics Education," in *Looking Back, Looking Forward: Proceedings of the Tenth International Conference on Teaching Statistics (ICOTS10, July, 2018)*, Kyoto, Japan: IASE, pp. 1–5.

Weins, A., Lovejoy, H. B., Mullen, Z., and Vance, E. A. (2022), "A Modelling Strategy to Estimate Conditional Probabilities of African Origins: The Collapse of the Oyo Empire and the Transatlantic Slave Trade, 1817–1836," *Journal of the Royal Statistical Society. Series A (Statistics in Society)*, 1–24. https://rss.onlinelibrary.wiley.com/doi/full/10.1111/rssa.12833. DOI:10.1111/rssa.12833